Gautham Manuru Prabhu

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Education

Manipal Institute of Technology, Manipal 2020 – 2024
B.Tech. in Computer Science and Engineering *CGPA:* ***8.91/10***

Work Experience

Cisco Systems Jan 2024 – Present
Software Engineering Intern → Software Engineer I → Software Engineer II *Bengaluru, India*

Software Engineer II, AI Acceleration Team *Oct 2025 – Present*

- Lead developer of an agentic test-triage platform for **hardware test failures on factory test lines** (RAG, LLM diagnosis, MCP), deployed in factories worldwide. It works out whether a failure traces to the test script, the test environment or the product, and finds root causes for **about 70% of previously undiagnosed failures**.
- Turned test and quality engineers' manual triage steps into agent workflows, and took the platform from prototype to production with the security, platform, infrastructure and transformation teams.
- Contributed to a support-queue agent that resolves **30% of routine requests**. Both systems cut case volume **35%** and MTTR **40%**, saving **$450,000** a year; presented to Cisco's Chief Supply Chain Officer.
- Mentor **5+ junior engineers** on software engineering, agent design, code review and operations. Received **25+ Cisco Connected Recognition awards**.

Software Engineer I, Supply Chain Transformation *Aug 2024 – Sep 2025*

- Deployed a **computer-vision inspection system** on assembly lines in multiple global factories that flags edge gaps in connectors from camera images, replacing manual visual checks.
- Added agentic AI to **failure analysis on factory test lines** and built a failure-classification suite that groups test failures by likely cause before engineers review them.
- Built Quality Transformation metrics engines tracking **10+ quality metrics** across **200+ product lines**.
- Split legacy monoliths into **10+ FastAPI microservices** on Kubernetes across **8 global sites**.

Software Engineering Intern, Supply Chain Organization *Jan 2024 – Jun 2024*

- Rebuilt a **Python 2** workflow as microservices with CI/CD, cutting deployment from **6 hours to 4 minutes**.
- Built a supplier-name standardization pipeline that improved consistency across **50,000+ supplier records**.

Hardware and Signal Processing

Project Kalpana: Student Radio Telescope, MIT Manipal 2022 – 2023
Co-founder and Technical Head *Manipal*

- Co-founded a student group building a radio telescope; my proposal won the **$13,000** university grant.
- Led the **software-defined radio** and **signal processing** work.

Research Experience

MiCoSyS Lab, San José State University March 2026 – Present
Research Intern | Advisor: Dr. Saptarshi Sengupta *Remote*

- Tested whether LLM-generated semantics improve temporal link prediction on DTGB text-attributed graphs (Enron, ICEWS1819) once all models are evaluated under one common protocol.
- Reproduced CROSS, a reported state-of-the-art LLM-augmented model, and matched its published results; compared it with DyGFormer and a lightweight structure-aware gate (SAGE) under a **100-negative** protocol over **5 seeds**.
- Found the gains depend on protocol and dataset: CROSS helped most on unseen nodes but turned seed-dependent on ICEWS1819, SAGE matched it, and larger text encoders did not help.
- **First-author poster** accepted at the Stanford Graph Learning Workshop (SNAP) [W1]; A* conference manuscript in preparation.

Medical Informatics Lab, IIT Kharagpur Summer 2023
Research Intern | Advisor: Dr. Subhamoy Mandal *Remote*

- Built dual-encoder, cross-attention vision models to detect age-related macular degeneration from **thermal and structural OCT**, robust to missing modalities and inconsistent resolutions.
- Studied image fusion across RGB, HSV and thermal representations; presented at the Digital Health Symposium, IIT Kharagpur [T1].

Cybersecurity and Quantum Computing Group, MIT Manipal 2022 – 2023
Undergraduate Researcher | Advisor: Dr. Vivekananda Bhat K. *Manipal*

- Benchmarked QSVC, Pegasos QSVC and quanvolutional networks for ECG-image cardiovascular classification; the latter reached **97.31% accuracy**, **3.88 percentage points** above a comparable CNN.
- Extended Pegasos QSVC from binary to four-class classification with six one-vs-one models; tuned **feature maps, qubit counts and circuit depth**. Supported by a **$10,200 MeitY–AWS QCAL grant**; published in *IEEE Access* [1].

Biometrics and Software Engineering Group, MIT Manipal 2022 – 2023
Undergraduate Researcher | Advisor: Dr. Srikanth Prabhu *Manipal*

- Co-developed EyeEncrypt, secure retinal vessel segmentation for **3,000+ fundus images** (Diffie–Hellman, AES-256); demonstrated it to Kasturba Medical College's ophthalmologists [4, 5].

Department of Information and Communication Technology, MIT Manipal 2022 – 2023
Research Intern | Advisor: Dr. Nisha P. Shetty *Manipal*

- Fine-tuned and compared BERT and XLNet on the **15,073-tweet** COVID-19 ANTi-Vax dataset, and used NetworkX to find misinformation clusters among Reddit authors [2].

Publications

Six peer-reviewed publications and one accepted workshop poster; 60+ citations (Google Scholar).

[1] S. Prabhu, S. Gupta, **G. M. Prabhu**, A. V. Dhanuka, K. V. Bhat. "QuCardio: Application of quantum machine learning for detection of cardiovascular diseases." *IEEE Access*, vol. 11, pp. 136122–136135, 2023. [doi]

[2] C. Rao, **G. M. Prabhu**, A. R. Kumar, S. Gupta, N. P. Shetty. "Addressing vaccine misinformation on social media by leveraging transformers and user association dynamics." *Procedia Computer Science*, vol. 235, 2024. [doi]

[3] **G. M. Prabhu**, T. Gupta, M. V. Srujan, A. R. Soumya, A. Palorkar, A. Chowdhury. "VIKAS: A multimodal framework to aid in effective disaster management." *ATIS 2022*, Springer CCIS, vol. 1804, 2023. [doi]

[4] G. Hegde, S. Gupta, **G. M. Prabhu**, S. V. Bhandary. "EyeEncrypt: A cyber-secured framework for retinal image segmentation." *ATIS 2022*, Springer CCIS, vol. 1804, 2023. [doi]

[5] G. Hegde, S. Prabhu, S. Gupta, **G. M. Prabhu**, A. Palorkar, M. V. Srujan, S. V. Bhandary. "A systematic review of deep learning approaches for vessel segmentation in retinal fundus images." *Journal of Physics: Conference Series*, 2023. [doi]

[6] A. A. Iyer, **G. M. Prabhu**, T. Gupta, S. Deshmukh, R. Rivankar. "Exploring design optimisation techniques of a radio telescope using fixed costing constraints." *Journal of Physics: Conference Series*, 2023. [doi]

[W1] **G. M. Prabhu**, S. Gupta, M. Choudhary, A. Nagarajan, K. Potika, S. Sengupta. "On link prediction using node-pair LLM semantics and structure in temporal text-attributed graphs." *Stanford Graph Learning Workshop (SNAP)*, 2026 (**first-author** poster).

Talks, Summer Schools, and Workshops

[T1] **G. M. Prabhu**, S. Gupta, S. Rathinam. "Enhanced analytics and procedural decision-making for ocular diagnostics using multi-modal data." *Digital Health Symposium*, IIT Kharagpur, 2023.

[T2] **G. M. Prabhu**, S. Gupta. "SatelTensor: Satellite data exploration via tensor decomposition." *TCML*, IISc, 2023.

[S1] ACM Winter School on Optimization for Machine Learning and Operations Research, IIT Goa, 2022.

[S2] Summer School on Dynamic Resource Allocation, Centre for Networked Intelligence, IISc Bengaluru, 2023.

Honors and Achievements

Winner, Offensive Summit Capture the Flag (CTF) security competition, Cisco 2026
25+ Cisco Connected Recognition awards 2024 – 2026
Second runner-up, Cisco Intern Case Study Competition 2024
Reviewer, International Conference on Machine Learning and Data Engineering (ICMLDE) 2025
Undergraduate Research Sponsorship Grant, Manipal Academy of Higher Education 2023
$10,200 MeitY–AWS Quantum Computing Applications Lab (QCAL) research grant for QuCardio [1] 2023
Grand finalist, Global Quantum Science and Technology Hackathon (top **16** of **1,626** teams) 2022
Grand finalist and team lead, Smart India Hackathon (**160,000+** students, **2,235+** institutes) [3] 2022
NTSE State Scholar, Government of Karnataka (rank **21** among **151,000+** candidates) 2018

Leadership

Lead Volunteer, Cisco Emerging Talent Recruitment, University Connect Track 2026 – Present
Track Lead Volunteer, Cisco Global Supply Chain, Innovation Track 2026 – Present
General Secretary and Treasurer, ACM Manipal Student Chapter 2022 – 2023
Technical Head, The Astronomy Club of Manipal 2021 – 2023

Skills and Interests

Hardware & Test	Hardware test triage, failure analysis on factory test lines, computer-vision inspection, software-defined radio, signal processing, quantum circuits
Languages	Python, C++, SQL, Bash
ML & Vision	PyTorch, PyG, DGL, Hugging Face, OpenCV, Qiskit, NetworkX
LLM & Agents	LangChain, LangGraph, LangSmith, MCP, Ragas, DeepEval, Pinecone, Weaviate
Systems & Cloud	FastAPI, microservices, Docker, Kubernetes, AWS, GCP, Jenkins, GitHub Actions, Git
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Neo4j
Research	Graph ML, temporal GNNs, LLMs, PEFT (LoRA), RAG, agentic systems, quantum ML
Interests	Volunteering, scuba diving, quizzing, trekking, astronomy